

# UPPER EXTREMITY FRACTURES — PART 1 (Shoulder Girdle → Elbow)

## Clavicle

- Fall on shoulder/outstretched hand. **Groups: I middle third (commonest), II lateral third, III medial third.**
- **Non-operative:** <2cm shortening/displacement, no neurovascular injury.
- **Operative (absolute):** open fracture, skin tenting, neurovascular injury, floating shoulder (clavicle + scapular neck), symptomatic non/malunion.

## Scapula

- Body fractures → no surgery. Glenoid neck (impacted) → conservative. **Intra-articular → surgery.** Acromion (undisplaced → non-op). Coracoid (proximal to ligaments → may need surgery, often with AC separation).

## Scapulothoracic dissociation

- **High-energy** — scapula/arm wrenched from chest → **rupture subclavian vessels + brachial plexus.** Flail ischaemic limb; **>1cm clavicle distraction = suspicion. Poor outcome** (reconstruction unlikely to help).

## AC joint injuries

- Direct trauma. **AC ligament** = anterior/posterior stability; **CC ligaments (trapezoid + conoid)** = superior/inferior stability.

## Sternoclavicular dislocation

- Lateral shoulder compression. **Anterior** = bump, painful, no cardiothoracic issue. **Posterior (rare, serious)** = tracheal/vessel pressure, venous congestion. **Serendipity view** (40° cephalic tilt).

## Shoulder dislocation

- **Anterior (commonest):** fall on hand → head forward → **Bankart lesion** (glenoid labrum avulsion).
- **Posterior:** severe internal rotation/adduction — **fit/convulsion/electric shock.**
- **Inferior (luxatio erecta):** arm fixed in abduction, humeral head into axilla.
- **Reduction:** Kocher's, Hippocratic, Stimson's, Milch's.
- **Complications:** early — rotator cuff tear, **axillary nerve injury (commonest)**, axillary artery, fracture-dislocation; late — stiffness (>40yrs), unreduced, **recurrent dislocation.**

## Proximal humerus fractures

- After middle age; **osteoporotic postmenopausal women;** fall on outstretched arm.
- **Minimally displaced (majority):** sling + early passive movements; active at 6 weeks.
- **Neer parts:** 2-part (surgical neck, greater tuberosity, anatomical neck), 3-part, **4-part (high complication — vascular, brachial plexus, AVN).**
- **Complications:** axillary nerve, AVN, stiffness, malunion.

- **Children (Salter-Harris):** SH-I <5yrs, SH-II >12yrs; metaphyseal 5–12yrs. Mostly non-operative (remodelling). Acceptable angulation: <10yrs any; 10–12yrs 60–75°; >12yrs 45°/■ displacement. Operative if soft tissue interposition (biceps tendon), open, vascular, intra-articular.

## Distal humerus (adults) — AO classification

- Often high-energy + nerve/vascular damage; elbow stiffness tendency. **Type A** extra-articular supracondylar; **Type B** unicondylar; **Type C** bicondylar.
- Supracondylar (rare adults) → ORIF. Type B/C displaced → ORIF (CT for planning); undisplaced → posterior slab. Alternatives: elbow replacement, "bag of bones", traction.

## Capitulum fracture

- Adults; fall on hand, elbow straight. Fullness anterior elbow. **Bryan-Morrey:** I complete, II cartilaginous shell, III comminuted. Displaced → operative.

## Radial head fracture

- Fall on outstretched hand, forearm pronated → impaction against capitulum. Tender radial head, pain on pronation/supination (**often missed**).
- **Mason:** I undisplaced split, II displaced single fragment, III comminuted, IV + elbow dislocation.
- Tx: I → collar and cuff 3wks; II → screw fixation; III → challenging. Complications: stiffness, myositis ossificans, instability (if MCL injured + head excised).

## Olecranon fracture

- Comminuted (direct blow) or **transverse (traction, triceps contraction)**. Check radial head (may be dislocated). Displaced transverse → operative (tension band).

## Elbow dislocation

- **90% posterior/posterolateral** — fall on outstretched hand, elbow extended. Check vascular/nerve (brachial artery, median/ulnar). Complications: stiffness.

## Paediatric elbow fractures

- 2nd commonest after distal forearm. **Supracondylar = commonest.**
- **95% posterior displacement** (hyperextension, fall on outstretched hand). Proximal fragment → **brachial artery / median nerve.**
- **Gartland:** I undisplaced, II angulated (posterior cortex intact), III completely displaced.
- Check pulse + capillary return; **fat pad sign** (undisplaced). Tx: I → cast 90°; II/III → reduction + **percutaneous crossed K-wires.**
- Complications: vascular, **nerve (radial/median/AIN/ulnar)**, **malunion (gunstock deformity)**, stiffness/myositis ossificans.
- **Lateral condyle** (2nd commonest paediatric): **higher risk of non-union, malunion, AVN.** Displaced → ORIF.
- **Medial epicondyle:** avulsion; surgery if entrapped in joint, articular, open.

## High-yield one-liners

- **Clavicle:** group I middle third commonest; floating shoulder/open/neurovascular = surgery.
- **Anterior shoulder dislocation = Bankart lesion + axillary nerve injury; posterior = fit/shock.**

- **Proximal humerus = osteoporotic women; 4-part = AVN risk; axillary nerve at risk.**
- **Supracondylar (children) = commonest paediatric elbow, 95% posterior, Gartland, brachial artery/median nerve, crossed K-wires, gunstock deformity.**
- **Radial head = Mason; olecranon transverse = tension band; elbow dislocation 90% posterior.**

# UPPER EXTREMITY FRACTURES PART 1 — Revision Table

*Bold = high-yield. Shoulder girdle → elbow. Dr Zaid Al-Bachachy.*

## SHOULDER GIRDLE

| Injury                       | Key points                                                                                                                                         |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Clavicle                     | Groups: <b>I middle (commonest)</b> , <b>II lateral</b> , <b>III medial</b> . Surgery: open, skin tenting, neurovascular, <b>floating shoulder</b> |
| Scapula                      | Body/neck conservative; <b>intra-articular</b> → <b>surgery</b>                                                                                    |
| Scapulothoracic dissociation | High-energy; subclavian + brachial plexus rupture; <b>&gt;1cm clavicle distraction</b> ; poor outcome                                              |
| AC joint                     | <b>AC lig = AP stability; CC (trapezoid+conoid) = SI stability</b>                                                                                 |
| Sternoclavicular             | Anterior (benign bump) vs <b>posterior (serious — tracheal/vessel)</b> ; serendipity view                                                          |

## SHOULDER & HUMERUS

| Injury                           | Key points                                                                                                               |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Anterior dislocation (commonest) | Fall on hand → <b>Bankart lesion</b> ; <b>axillary nerve injury (commonest)</b>                                          |
| Posterior dislocation            | <b>Fit/convulsion/electric shock</b> (internal rotation/adduction)                                                       |
| Luxatio erecta                   | Inferior; arm fixed abducted                                                                                             |
| Proximal humerus                 | <b>Osteoporotic postmenopausal women</b> ; Neer parts; <b>4-part = AVN risk</b> ; minimally displaced (majority) → sling |
| Paediatric proximal              | Salter-Harris; mostly non-operative (remodelling)                                                                        |

## ELBOW & DISTAL HUMERUS

| Injury                     | Key points                                                                                                                                         |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Distal humerus (adult)     | <b>AO: A extra-articular, B unicondylar, C bicondylar</b> ; displaced → ORIF                                                                       |
| Radial head                | <b>Mason I–IV</b> ; missed often; pain on pronation/supination; II → screw                                                                         |
| Olecranon                  | Comminuted (blow) vs <b>transverse (traction)</b> → <b>tension band</b>                                                                            |
| Elbow dislocation          | <b>90% posterior</b> ; brachial artery, median/ulnar nerve                                                                                         |
| Supracondylar (children)   | <b>Commonest paediatric elbow; 95% posterior</b> ; Gartland I–III; brachial artery/median nerve; crossed K-wires; gunstock deformity; fat pad sign |
| Lateral condyle (children) | 2nd commonest; <b>non-union/malunion/AVN risk</b>                                                                                                  |